

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS student (NC State, May 2026) with production backend experience in Python, Node.js, and distributed systems across multiple deployed products. At Isomer AI, built backend pipelines for document processing (PDF/email/S3), RBAC systems, and multi-model AI integrations across 7 production deployments -- 394 commits to a live TypeScript/Node.js/PostgreSQL codebase. Built AlgoPulse, scaling an AI orchestration backend to 2,000 async inference requests/second using Temporal.io durable workflows and Kafka state sync. At Mphasis, built OCR/NLP/ML automation workflows in Python improving claim accuracy by 25%. Strong in API design, distributed pipelines, cloud infrastructure (AWS, GCP, Docker), and integrating AI research into production systems.

CORE COMPETENCIES

Python (Backend & ML)

Node.js / Express APIs

Distributed Systems (Kafka / Temporal)

PostgreSQL & Redis

Cloud Infrastructure (AWS / GCP / Docker)

AI Model Integration

REST API Design

Scalability & Performance

TypeScript & React

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Remote

- Built backend pipelines for **document processing (PDF generation, email parsing via Microsoft Graph API, S3 storage)**, RBAC system, and multi-model AI integrations across Claude, GPT-4o, and Gemini using MCP tool-calling -- **394 commits** across 7 production deployments
- Integrated Temporal.io for durable AI workflow orchestration across async reasoning loops; wired Auth0, Redis, Docker, and Sentry across the full TypeScript/Node.js/PostgreSQL stack
- Built real-time event pipelines (Pusher) for live status propagation; maintained zero-downtime across all production releases in a fast-moving startup environment

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built **5+ Python automation workflows** combining OCR, NLP, and ML classification to automate health insurance claim processing -- improved accuracy by **25%**
- Automated model validation and deployment pipelines on AWS & GCP; integrated Grafana dashboards for real-time model performance monitoring

NC State University Apr 2025 - Present

Graduate Research Assistant

- Built a cloud-based genomic data platform on **Google Cloud Run** serving 100+ researchers across 500GB of datasets; applied ML hyperparameter tuning improving model accuracy by 12%

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built **Kafka streaming + Spark batch pipelines** for real-time support ticket classification; reduced pipeline latency by 20% and processing time by 15%

PROJECTS

AlgoPulse - AI Orchestration Platform Independent · 2025

Identified a polling bottleneck under load and re-architected an AI platform to event-driven streaming: **Temporal.io** for durable workflow state, **Kafka** for worker coordination and fault tolerance, Supabase Edge Functions for real-time client streaming. Scaled to **2,000 async AI inference requests/second** with zero state loss under fault injection.

Python · TypeScript · Node.js · Temporal.io · Kafka · Supabase Edge Functions · PostgreSQL

Glance AI - Privacy-First RAG System Independent · 2025

Built a **compliance-ready Python RAG pipeline** for fintech and healthcare with per-tenant data isolation at the vector retrieval layer -- 0 cross-tenant context leakage by design. Deterministic retrieval architecture eliminates hallucinations for regulated data environments.

Python · LLMs · Vector Databases · RAG · Compliance-ready architecture

Mapify - ML & NLP Pipeline IEEE Published

T5-transformer pipeline in Python that converted 1,500+ PDFs and images into structured mind maps and flowcharts, reducing academic review time by 60%. Presented to 500+ attendees at IEEE conference.

Python · T5 Transformer · PyTorch · NLTK · NetworkX

PerCent - GenAI Finance App HackNC24 Top 10 of 100+ teams

Built full Python backend with RAG-powered AI chatbot (Qwen-14B-Chat via Hugging Face) and Selenium data ingestion pipeline reducing manual financial research by 30%. Complete MVP in 48 hours, top 10 at HackNC24.

Python · Hugging Face · Qwen-14B-Chat · RAG · Selenium · React

EDUCATION

North Carolina State University - MS Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University - BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure	

SKILLS

- Backend: Python, Node.js, Express, PostgreSQL, Redis, Prisma, REST APIs, Flask
- Distributed Systems: Kafka, Spark, Temporal.io, Hadoop, RabbitMQ-adjacent (Kafka)
- Cloud & Infra: AWS, GCP, Docker, Kubernetes, CI/CD, Grafana, Sentry, Google Cloud Run
- AI/ML: PyTorch, TensorFlow, LangChain, MCP, RAG, OpenAI API, Anthropic Claude, Hugging Face, Pandas, NumPy
- Frontend: React, TypeScript, JavaScript (ES6+), Next.js, HTML/CSS